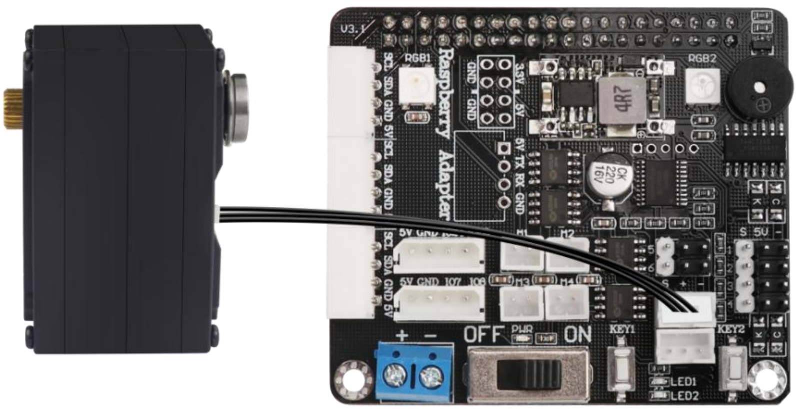
Lesson 1 Control Bus Servo Rotate

1. Preparation

1.1 Hardware wiring

Connect bus servo to any bus servo port of Raspberry Pi board separately. Here we take LX-15D as an example.



Note: the bus servo wire adopts anti-reverse plug design, please insert carefully.

1.2 ID Set

We will take the example of controlling ID6 servo. It can be set through the "Servo Test" in the ArmPi FPV PC computer.

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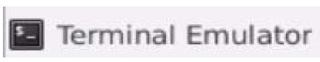
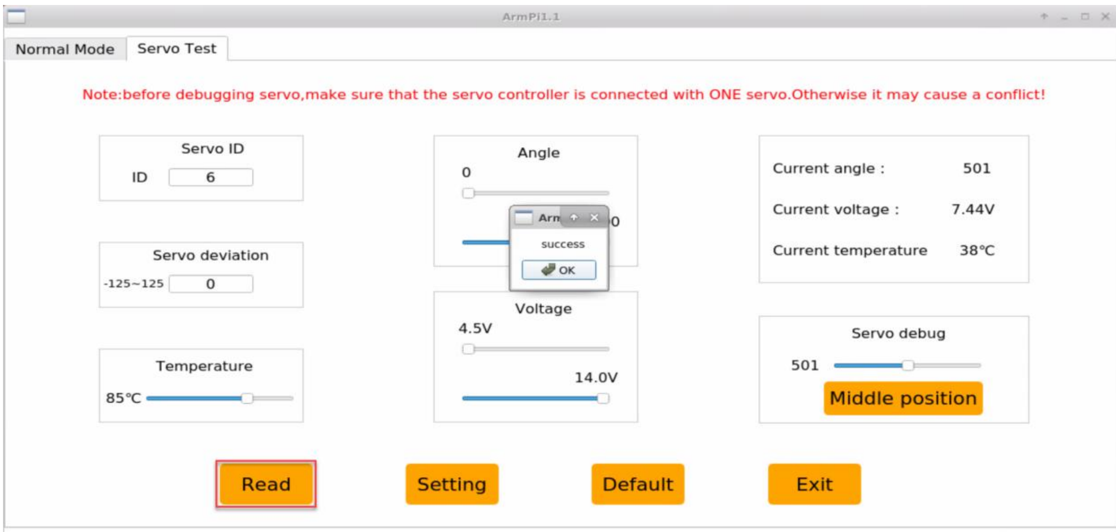
1) Open LX terminal. Click 

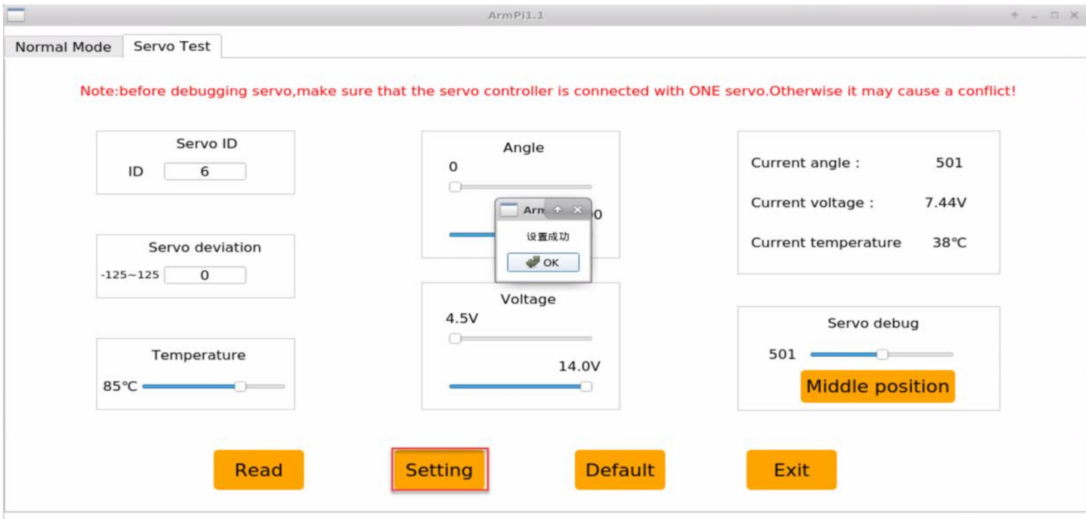
and select 

to open command line terminal of Ubuntu. Then enter “ sudo ArmPi_PC_Software/ArmPi.py” command in the terminal window to open the PC software.

2) Click and switch to the "Servo Test" menu bar, then click “Read” and wait until the “success” prompt occurs.



3) In the "Servo ID" area, select the ID, input servo number 6, click "Set" button, and wait for the prompt "success".



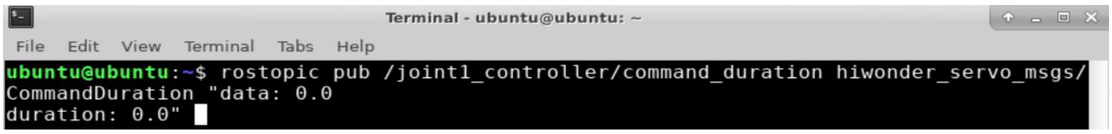
2. Operation Steps

1) Open LX terminal. Click 

and select 

to open command line terminal of Ubuntu.

2) Initially, enter "**rostopic pub /joint1_controller/command_duration**" command, then complete the command through TAN key.



3) Next, revise parameters of "data" and "duration" in the command. For example, we modify the parameter of data as 1.5 and duration as 500. Then press Enter.

Note: data refers to the rotation range of servo, from -2rad to 2rad. Duration is for rotation time in milliseconds.

4) If you want to quit this program, press “Ctrl+C”.

3. Project Outcome

The servo connected to the Raspberry Pi expansion board will rotate to designed angle.